

6.3 Gravitation

Kepler formulated the laws of rotational motion of planets revolving around the sun. But Kepler did not have any idea about the influencing force that revolves the planets around sun. Newton inspired from the laws of planetary motion started thinking about the forces between them. There is a story like that, one day while sitting in the garden Newton saw an apple to fall in the ground. The question aroused in his mind "why does the apple fall in the ground?" Certainly somebody is pulling it towards the ground. From these thoughts he came to the conclusion that the earth attracts everybody towards itself. Not only the earth, but as a reason of revolution of planets around the sun he imagined a force between two bodies. He said in this universe everybody attracts every other body towards itself. The magnitude of this force depends only on the masses of the bodies and the distance between them—but does not depend on the size and shape, nature, direction and medium. The attraction between any two bodies of this universe is called the gravitation.

Definition : The mutual attraction between two bodies of this universe is called gravitation.

6.4 Newton's Law of Gravitation

In this universe everybody attracts every other body towards itself. This attraction is called the gravitation. Regarding this attraction Newton formulated a law which is known as Newton's law of gravitation.

Law : Every particle in the universe attracts each other with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them and the force acts along the straight line joining the two particles.

Explanation : Let m_1 and m_2 be the masses of two bodies separated by a distance d (Fig. 6.3)

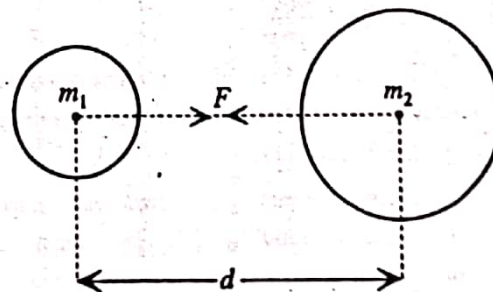


Fig 6.3

If F be force of attraction between them, then according to gravitational law,

$$F \propto m_1 m_2, \text{ when } d \text{ remains constant, and}$$

$$F \propto \frac{1}{d^2}, \text{ when } m_1 \text{ and } m_2 \text{ remain constant.}$$

$\therefore$ According to the law of proportionality,

$$F = \frac{G m_1 m_2}{d^2}$$

Gravitation is referred to the force acting between two bodies which can be represented as $f = \frac{G m_1 m_2}{d^2}$

$F \propto \frac{m_1 m_2}{d^2}$, when m_1 , m_2 and d all change.

or, $F = G \frac{m_1 m_2}{d^2}$... (6.1)

Here G is proportionality constant. It is called universal gravitational constant.

6.5 Gravitational Constant, G

Definition of G

From equation (6.1), we see that, when $m_1 = m_2 = 1$ unit and $d = 1$ unit

then $F = G \frac{1 \times 1}{1^2}$ i.e; $F = G$ or, $G = F$

Definition : The magnitude of the force of attraction between two bodies of unit masses placed at unit distance is called the gravitational constant.

Dimension of G

From eq. (6.1), we see that, $G = \frac{Fd^2}{m_1 m_2}$ or, $G = \frac{\text{Force} \times (\text{distance})^2}{(\text{mass})^2}$

So, the dimension of G would be the dimension of the right hand sided quantities. That is

$$[G] = \frac{MLT^{-2} \times L^2}{M^2} = L^3 M^{-1} T^{-2}$$

Unit of G

Again from eq. (6.1), we see that,

$$G = \frac{Fd^2}{m_1 m_2} \quad \dots \quad \dots \quad \dots \quad (6.2)$$

If we put the units of right hand sided quantities, then we get the unit of G . In international system, the unit of G is Nm^2kg^{-2} or, $m^3kg^{-1}s^{-2}$

Magnitude of G

Many scientists performed many experiments to determine the magnitude of G . The magnitudes found by different scientists slightly vary with each other. The accepted magnitude of G in international system is

$$6.673 \times 10^{-11} Nm^2kg^{-2}$$

Significance of G

The magnitude of G is $6.673 \times 10^{-11} Nm^2kg^{-2}$ means if two bodies each of masses 1 kg are placed at a distance 1m apart, then they will attract each other with $6.673 \times 10^{-11}N$ force.

Universality of G

Gravitational constant G is a universal constant, because the gravitational force does not depend on the medium between two objects. Its magnitude does not depend on any property

of the medium between two objects such as permeability, susceptibility or directionality of the medium.

Again, the gravitational force depends on the masses of the objects, but not on the materials that made the objects. That is, there is no relation of G with the natures of the objects.

6.6 Kepler's Laws of Planetary Motion

In ancient time, the Greeks used to think about solar system. Ptolemy presented the observation of Greek astronomers as a theory. According to him, the earth is the centre of the universe and the sun, moon, planets and stars move around the earth in complex orbits. Though the theory of Ptolemy prevailed up to fifteenth century, but this theory was very complex and was unable to explain many observations. In sixteenth century, Copernicus said, if sun is considered the fixed centre of the universe, then a simple explanation of the heavenly bodies can be given. According to the theory of Copernicus, earth is a planet which revolves round its axis and rotates around the sun and same type of motion are also present in case of other planets like earth.

The conflict between these two theories made astronomers more interested to collect more observed information. Tyco Brahe collected these information. His colleague Johan Kepler analyzed and explained these information and found that there was regularity in the motion of the planets. These regularities are known as three laws of planetary motion of Kepler.

Kepler's Laws :

First Law : Law of Orbit : Each planet moves in a (elliptical) orbit with the (sun) at one focus.

Second Law : Law of Area : Each planet moves in such a way that the line joining the sun and the planet sweeps out equal areas in equal times.

Third Law : Law of Time Period : The square of the period of revolution of any planet around the sun is proportional to the cube of the semi major axis of its orbit.

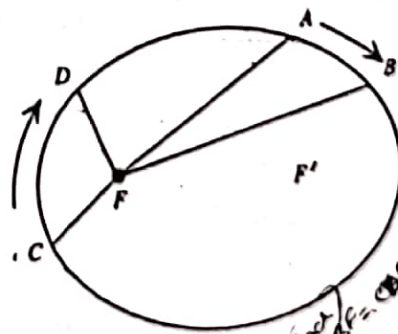


Fig. 6.4

$$(Time)^2 \propto (semi\ major\ axis)^3$$

$T^2 \propto a^3$ distance

Explanation of Laws

First Law : In fig. 6.4, $ABCD$ is an elliptical orbit. F and F' are two foci of this ellipse. According to Kepler's first law the sun will be at any focus and the planet will revolve in elliptical orbit.

Second Law : Suppose in fig 6.4, the sun is at focus F . If a planet in the orbit moves from position A to position B in time t and if it also takes same time t to move from position C to position D , then according to Kepler's second law the area AFB will be equal to area CFD .

Third Law : The planets revolve the sun in elliptical orbits. Thus the distance of a planet from sun varies with time. Let a be the semi major axis of the orbit of a planet and T be the time taken for one revolution. According to third law of Kepler,

$$T^2 \propto a^3$$

i.e., if the time periods of the planets whose semi major axes of the orbits are a_1, a_2, a_3 from the sun be T_1, T_2, T_3 respectively then

$$\frac{T_1^2}{a_1^3} = \frac{T_2^2}{a_2^3} = \frac{T_3^2}{a_3^3} = \text{constant}$$

6.7 Deduction of Mathematical Expression of Kepler's law from Newton's law of Gravitation

Let the planet P of mass m is revolving around the sun in a circular orbit of radius r (fig.5).

The gravitational force between the sun and the planet is,

$$F_G = \frac{GMm}{r^2}$$

Here M is the mass of the sun.

The planet is revolving in a circular orbit.

So the centripetal force is,

$$F_c = \frac{mv^2}{r}$$

Here v is the orbital speed of the planet.

To revolve the planet in a circular orbit the gravitation force and the centripetal force should be equal, so,

$$F_G = F_c$$

$$\text{or, } \frac{GMm}{r^2} = \frac{mv^2}{r}$$

The time period of the planet around the sun is T .

(6.3)

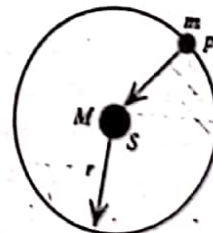


Fig 6.5

$$\therefore T = \frac{2\pi r}{v}$$

$$\text{or, } v = \frac{2\pi r}{T}$$

Putting the value of v in equation (6.3) we get.

$$\frac{GMm}{r^2} = \frac{m}{r} \left(\frac{2\pi r}{T} \right)^2$$

$$\text{or, } \frac{GMm}{r^2} = \frac{m}{r} \frac{4\pi^2 r^2}{T^2}$$

$$\text{or, } T^2 = \frac{4\pi^2}{GM} r^3$$

$$\text{or, } T^2 = Kr^3$$

Here, $K = \frac{4\pi^2}{GM}$ is a constant.

$$\text{So, } T^2 \propto r^3$$

Which is the third law of Kepler.

The magnitude of K for each planet is same and this magnitude is $2.97 \times 10^{-19} \text{ s}^2 \text{m}^{-3}$. If the orbit of the planet is elliptical then instead of radius r we are to write semi major axis a .

6.8 Gravity and Acceleration due to gravity

Gravity : The attraction between two bodies of the universe is called gravitation. If one of the bodies is earth, then the attraction is called gravity i.e., the attraction of the earth on a body is called gravity.

Definition : The force of attraction between the earth and any other body is called gravity.

The attraction between the sun and the moon is gravitation but the attraction between the earth and a book is gravity.

Acceleration due to gravity : From Newton's second law of motion, we know that a force produces an acceleration of a body. Hence, due to the influence of the force of gravity a body will accelerate. This acceleration i.e., the rate of increase of velocity is called acceleration due to gravity.

Definition : The rate of change of velocity of a freely falling body under the influence of gravity is called acceleration due to gravity.

It is denoted by g . Since, acceleration due to gravity is one type of acceleration, its dimension and unit will be LT^{-2} and ms^{-2} respectively.

Equation of g

Let M = mass of the earth, m = mass of a body at earth's surface or near it, d = distance between body and the centre of the earth.

According to Newton's law of gravitation, we find

$$\text{force of gravity, } F = G \frac{Mm}{d^2} \quad \dots$$

But, from Newton's second law of motion, we find,

Force of gravity = mass $\times$ acceleration due to gravity.

$$\text{i.e., } F = mg \quad \dots$$

From eq. (6.4) and (6.5), we find $mg = \frac{GMm}{d^2}$ (6.5)

$$\text{or, } g = \frac{GM}{d^2} \quad \dots$$

We see that the mass of the body m is absent in the right hand side of the equation (7.4). (6.6)

Hence, acceleration due to gravity does not depend on the mass of the body. Since, G and mass of the earth M are constants, so the magnitude of g depends on the distance of the body from the centre of the earth. Thus, though the magnitude of g is independent of the body but it is not independent of the place.

Magnitude of g at earth's surface

If the distance of the earth's surface from the centre of the earth i.e., the earth's radius is R , then at earth's surface, $g = \frac{GM}{R^2}$ (6.7)

Since, the earth is not a perfect sphere, the earth's radius R is also not a constant. At the equatorial region the magnitude of R is maximum, hence g is minimum at this region and its magnitude is 9.78039 ms^{-2} . On the other hand, at the polar region R is minimum, hence g is maximum and its magnitude is 9.83217 ms^{-2} .

6.9 Variation of Acceleration due to Gravity.

The acceleration due to gravity is not a constant quantity. The value of g is different at different places. Below we describe his reasons of variation of g .

(b) Due to earth's shape : As the earth is not a perfect sphere, hence all the places at earth's surface are not at equal distance from the centre of the earth. Since the value of g depends on the distance from the centre of the earth hence g changes from place to place on earth's surface. The earth's radius is greatest at equatorial region, so g is minimum there and the value is 9.78039 ms^{-2} . The radius decreases towards polar region from equatorial region hence, g increases. As radius is minimum at polar region, so g is maximum there and the value is 9.83217 ms^{-2} . The magnitude of g at 45° latitude at sea level is taken as the standard

$$= 9 \times 6.4 \times 10^6 \text{ m}$$

$$= 57.6 \times 10^6 \text{ m}$$

$$\text{Ans } 57.6 \times 10^6 \text{ m}$$

Numerical Example 6.5. The ratio of radius of the Venus and Earth's orbit around the sun is 54 : 75. In the earth if a year consists of 365 days, how many days will form a year in the Venus?

Solution : We know,

$$\frac{T_1^2}{R_1^3} = \frac{T_2^2}{R_2^3}$$

$$\text{or, } T_1^2 = \left(\frac{R_1}{R_2}\right)^3 T_2^2$$

$$\text{or, } T_1^2 = \left(\frac{54}{75}\right)^3 \times (365 \text{ d})^2$$

$$\therefore T_1 = 223 \text{ d}$$

$$\text{Ans. } 223 \text{ d}$$

Here,

Time period of Venus, $T_1 = ?$

Time period of Earth, $T_2 = 365 \text{ d}$

If the radii of the Venus and Earth are

R_1 and R_2 respectively, then

$$\frac{R_1}{R_2} = \frac{54}{75}$$

6.11 Gravitational Field

Every body in this universe attracts each other towards themselves. In the neighbourhood of a body the space where this gravitational influence of the body exists is called its gravitational field. Theoretically the gravitational field extends up to infinity.

Definition : The space around a body where any other object experiences a gravitational attraction is called the gravitational field of the body.

Gravitational Field Strength or Gravitational Intensity

In the gravitational field of a body its influence is not same everywhere. Its influence is different at different points. This influence is measured by gravitational field strength or gravitational intensity. The gravitational field intensity is measured by the force that acts on a test body placed at different places in the gravitational field. An object of unit mass is taken as the test body. The gravitational field intensity is measured by the force experienced by this unit mass.

Definition : The gravitational force experienced by an object of unit mass placed at a point in the gravitational field is called the gravitational field intensity of the point.

Magnitude : If a body of mass m placed at a point in a gravitational field experiences a force F , then the force that will act on an unit mass at that point would be $\frac{F}{m}$. Hence, the gravitational field intensity,

$$E_G = \frac{F}{m}$$

...

...

...

(6.15)

$$\text{or, } F = mE_G$$

(6.16)

Thus, if we place an object of mass m at a point where the gravitational field intensity is E_G , then the force mE_G will act on it.

Direction : Since, gravitational field intensity is the force acting on unit mass, it has direction and hence, it is a vector quantity. The direction of gravitational intensity would be direction in which the unit mass experiences the force.

$$\vec{E}_G = \frac{\vec{F}}{m}$$

$$\text{and, } \vec{F} = m\vec{E}_G$$

(6.15a)

(6.16a)

Unit : From eq. (6.15), it is seen that by dividing the unit of force by unit of mass, unit of gravitational intensity is obtained. Hence, the unit is Nkg^{-1}

Significance : The gravitational intensity at a point is 5 Nkg^{-1} means if we place an object of mass 1 kg at that point, then it will experience 5 N force. If we place a 3 kg object there, then it will experience $3 \text{ kg} \times 5 \text{ Nkg}^{-1} = 15 \text{ N}$.

Gravitational Potential

The gravitational potential at a point in a gravitational field means the work done by gravitational force to bring a body of unit mass from infinity to that point.

Definition : The amount of work done by the gravitational force in bringing an object of a unit mass from infinity to a point in a gravitational field is called the gravitational potential of that point.

Magnitude : If the work done to bring a body of mass m from infinity to a point in the gravitational field is W ; then the gravitational potential at the point would be

$$V = \frac{W}{m}$$

(6.17)

The gravitational potential at infinity is taken as zero.

Direction : Since potential is work/mass and as work has no direction, so gravitational potential does not have any direction. It is a scalar quantity.

Dimension : From eq. (6.17) we find,

$$\text{Potential} = \frac{\text{work}}{\text{mass}}$$

Thus, dividing the dimension of work by the dimension of mass, we get the dimension of V .

$$[V] = \frac{ML^2T^{-2}}{M} = L^2T^{-2}$$

Unit : Dividing the unit of work by the unit of mass, we get the unit of gravitational potential. Thus, the unit of gravitational potential is Jkg^{-1} .

Relation between Gravitational Field Intensity and Gravitational Potential

Let a point mass M is placed at A . B is a point within the gravitational field of the mass M and at a distance r from it (fig 6.12). We connect AB . If we make a unit mass at B , then the gravitational force acting upon it will be,

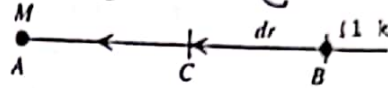


Fig. 6.12

$$E = G \frac{M \times 1}{r^2} = \frac{GM}{r^2}$$

Its direction would be along BA .

Suppose the potential at point B is V . Now if this unit mass moves an infinitesimal small distance dr to C towards the attractive force E exerted on it by the body of mass M , then the work done would be $E dr \cos 0^\circ$ and the change in potential dV would also be this. Thus

$$dV = E dr \cos 0^\circ$$

$$\therefore dV = E dr$$

$$\therefore E = \frac{dV}{dr}$$

Since the gravitational potential is taken to be maximum at infinity and this maximum value is Zero so proceeding from infinity towards the body which creates the gravitational field, gravitational potential decreases. That is it becomes negative and the value of gravitational intensity increases. So the above equation will be

$$E = - \frac{dV}{dr} \quad \dots \quad (6.18)$$

That is the rate of decrease of gravitational potential with distance or the negative differentiation is the gravitational intensity.

Expression for Gravitational Field Intensity

Suppose a body of mass M is situated at point A (Fig. 6.13). A body of unit mass is placed at a point B within its gravitational field and at a distance r from mass M .

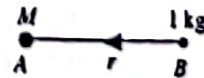


Fig 6.13

Now, according to Newton's law of gravitation, the force on

$$\text{this unit mass i.e., gravitational intensity is } E_G = G \frac{M \times 1}{r^2} = \frac{GM}{r^2} \quad \dots \quad (6.19)$$

and its direction would be along BA .

Gravitational Potential due to Point Mass

Let a point mass M is placed at A . We have to determine the gravitational potential at point B within the gravitational field of mass M and at a distance r from it (Fig. 6.14). We connect AB . If we place a unit mass at B , then the gravitational force acting upon it will be,

$$F = G \frac{M \times 1}{r^2} = \frac{GM}{r^2}$$

Its direction would be along BA . Suppose, the potential at point B is V . Now if this unit mass moves an infinitesimal small distance dr to C towards the attractive force F exerted on it by the body of mass M , then the work done would be $Fdr \cos 0^\circ$. The change in potential dV would also this. Thus,

$$dV = Fdr \cos 0^\circ$$

$$= Fdr$$

Now, integrating this equation from $r = \infty$ to $r = r$, we get the potential V at point B .

$$\begin{aligned} V &= \int_{\infty}^r Fdr \\ &= \int_{\infty}^r \frac{GM}{r^2} dr \\ &= GM \int_{\infty}^r \frac{dr}{r^2} \\ &= GM \left[-\frac{1}{r} \right]_{\infty}^r \\ &= GM \left[-\frac{1}{r} + \frac{1}{\infty} \right] \\ \text{or, } V &= -\frac{GM}{r} \end{aligned} \quad (6.20)$$

This negative sign indicates that gravitational potential would be maximum at infinity and this maximum value is zero. Proceeding from infinity towards the body which creates the gravitational field, gravitational potential decreases i.e., it becomes negative.

6.12 Application of law of Gravitation

Using law of gravitation we can determine the gravitational potential and gravitational intensity inside and outside of a sphere. We shall determine the gravitational potential and intensity of a uniform solid sphere.

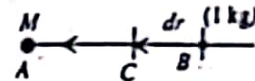


Fig. 6.14

compound pendulum: A compound pendulum has an extended mass, like a swinging bar, and is free to oscillate about a horizontal axis. A special reversible compound pendulum designed to measure the value of g , the acceleration of gravity.



∴ gravitational intensity at P

$$E = -\frac{dV}{dr} = -\frac{d}{dr} \left[-\frac{GM}{2a^3} (3a^2 - r^2) \right]$$

$$= \frac{GM}{2a^3} \frac{d}{dr} (3a^2 - r^2)$$

$$\therefore E = -\frac{GM}{a^3} r$$

(6.24)

Here the negative sign indicates attractive force. If we consider magnitude only, then

$$E = \frac{GM}{a^3} r$$

Numerical Example 6.6. The radius of the earth is 6.4×10^6 m; its mean density is $5.5 \times 10^3 \text{ kg m}^{-3}$. The value of $G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$. Considering the earth as a sphere determine the gravitational potential on its surface.

Solution : We know,

$$V = -\frac{GM}{R}$$

But here the mass of the Earth,

$$M = \frac{4}{3} \pi R^3 \rho$$

$$\therefore V = -\frac{G \frac{4}{3} \pi R^3 \rho}{R} = -\frac{4}{3} \pi G R^2 \rho$$

$$= -\frac{4}{3} \times 3.14 \times 6.7 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2} \times (6.4 \times 10^6 \text{ m})^2 \times 5.5 \times 10^3 \text{ kg m}^{-3}$$

$$= -6.32 \times 10^7 \text{ J kg}^{-1}$$

$$\text{Ans. } -6.32 \times 10^7 \text{ J kg}^{-1}$$

Here,

Radius, $R = 6.4 \times 10^6 \text{ m}$

Density, $\rho = 5.5 \times 10^3 \text{ kg m}^{-3}$

Potential, $V = ?$

Gravitational constant, $G = 6.7 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$

6.13 Escape Velocity

If you throw a stone upwards, it returns to earth's surface due to the gravitational attraction. But if we throw an object with such a velocity so that it can overcome the force of gravity, then the object would not return to earth. The minimum velocity with which an object is thrown upwards so that it can go to space overcoming earth's gravity is known as escape velocity.

Definition : The minimum velocity with which an object must be thrown vertically upwards so that it does not return to the earth is called the escape velocity.

Magnitude of Escape Velocity : An object must be given such a kinetic energy so that overcoming earth's gravity it can go to space. The same amount of work that has to be done to take an object from earth's surface to infinity is to be given to the object as kinetic energy. The velocity which is to be given to acquire this kinetic energy is escape velocity v_e .

Suppose, a body of mass m is thrown upwards with a velocity v_e . When the body reaches a distance r from earth's centre, then the gravitational attractive force it will experience towards earth's centre would be according to law of gravitation.

$$F = \frac{GMm}{r^2}$$

Here, M is the mass of earth. Now, to move this body an infinitesimal small distance dr against this force, the work dW that has to be done in expense of kinetic energy is

$$dW = Fdr$$

$$\text{or, } dW = \frac{GMm}{r^2} dr$$

Thus, the total work done to take this body from earth's surface (where $r = R$, earth's radius) to infinity (where $r = \infty$) would be

$$W = \int_R^{\infty} \frac{GMm}{r^2} dr$$

$$\text{or, } W = GMm \left[-\frac{1}{r} \right]_R^{\infty}$$

$$\text{or, } W = GMm \left[-\frac{1}{\infty} + \frac{1}{R} \right]$$

$$\text{or, } W = \frac{GMm}{R} \quad \dots \quad \dots \quad \dots \quad (6.25)$$

If by doing this amount of work the body has to go to space by overcoming earth's attraction, then at least this amount of kinetic energy has to be imparted to the body at the time of throw. For this kinetic energy if the body is to be given the velocity v_e , then

$$\frac{1}{2} mv_e^2 = \frac{GMm}{R}$$

$$\text{or, } v_e^2 = \frac{2GM}{R}$$

$$\text{or, } v_e = \sqrt{\frac{2GM}{R}} \quad \dots \quad \dots \quad \dots \quad (6.26)$$

Again, at earth's surface $g = \frac{GM}{R^2}$, hence this equation becomes

$$v_e = \sqrt{2gR} \quad \dots \quad \dots \quad \dots \quad (6.27)$$

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Though, equations (6.26) and (6.27) are deduced for escape velocity in case of earth; it will be applicable for any planet or satellite of this universe. In that case, g , R and M would be the acceleration due to gravity, radius and mass of the respective planet or satellite.

Magnitude of Escape Velocity at Earth

Radius of the earth, $R = 6.4 \times 10^6$ m and

at earth's surface, $g = 9.8 \text{ ms}^{-2}$

$$\begin{aligned} \text{Hence, } v_e &= \sqrt{2 \times 9.8 \text{ ms}^{-2} \times 6.4 \times 10^6 \text{ m}} \\ &= 11200 \text{ ms}^{-1} = 11.2 \text{ kms}^{-1} \end{aligned}$$

Thus, if a body is thrown vertically upwards with a minimum velocity 11.2 kms^{-1} , then it will go to space overcoming earth's attraction. From equation (6.21) or (6.22), we see that mass of the body m is absent in the expression of escape velocity. Thus, it is seen that whether the body is small or big, the same velocity is to be given to it in order to go to space forever overcoming earth's attraction.

6.14 Uses of law of Gravitation

Artificial Satellite

A main use of law of gravitation is to revolve artificial satellite around the world. The spaceship that revolve around the earth in a definite orbit are called artificial satellite they are projected with the help of rockets. For revolving around the earth, the artificial satellite needs centripetal force. The attraction of earth on the artificial satellite i.e. the force of gravity on it supplies this centripetal force. Due to attraction of gravity they revolve in their orbit like moon. To revolve in a particular orbit they require necessary speed. The speed required to revolve in a fixed orbit is given by

$$v = \sqrt{\frac{GM}{R+h}} \quad \dots \quad (6.28)$$

Here, G = Gravitational constant, M = Mass of the earth, R = radius of the earth, h = distance of the artificial satellite from the earth's surface.

The time period of the artificial satellite is

$$\begin{aligned} T &= \sqrt{\frac{4\pi^2(R+h)^3}{GM}} \\ \therefore T &= 2\pi \sqrt{\frac{(R+h)^3}{GM}} \quad \dots \quad (6.29) \end{aligned}$$

$$= F dr$$

$$= (GM/r^2) dr$$

Now, integrating this equation from $r = \infty$ to $r = r$, we get

$$W = \int_{\infty}^r (GM/r^2) dr$$

$$= - \frac{GM}{r}$$

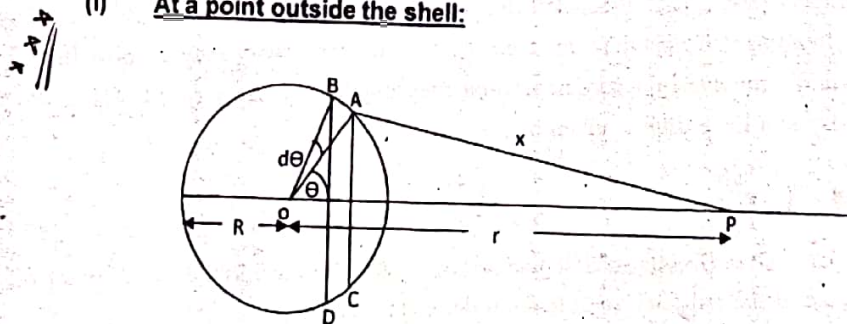
Hence, the potential at B is

$$V = W = - \frac{GM}{r}$$

The negative sign indicates that gravitational potential would be maximum at infinity and this maximum value is zero. Proceeding from infinity towards the body which creates the gravitational field, gravitational potential decreases.

Gravitational potential and field at a point due to a Spherical Shell:

(i) At a point outside the shell:



Consider a uniform spherical shell of mass M and radius R . Let ρ be the density per unit area of the shell. The planes AC and BD cut the shell vertically and the element between the two planes is a ring of radius $R \sin \theta$.

So the length of AB,

$$l = R d\theta$$

$$\text{Surface area of elements} = 2\pi R \sin \theta R d\theta$$

$$= 2\pi R^2 \sin \theta d\theta$$

Shell
is a sphere
Surface
or volume
mass
Surface
volume

Mass of the element,

$$m = (2\pi R^2 \sin\theta d\theta) \rho$$

If x is the distance of the point P from A, then the potential at P due to the element,

$$dV = -Gm/x$$

$$= -(G2\pi\rho R^2 \sin\theta d\theta) / x$$

From the triangle ΔOAP

$$x^2 = R^2 + r^2 - 2Rr \cos\theta$$

differentiating above equation we get,

$$x = (rR \sin\theta d\theta) / dx$$

So the potential,

$$dV = -(G2\pi\rho R^2 \sin\theta d\theta) dx / (rR \sin\theta d\theta)$$

$$= -(2\pi\rho GR / r) dx$$

Now integrating for the whole shell

$$V = - \int_{r-R}^r \frac{R2\pi\rho GR}{r} dx$$

$$= -(4\pi R^2 \rho G / r)$$

But the Mass of the sphere,

$$M = 4\pi R^2 \rho$$

$$\text{So, } V = -GM / r$$

Thus for a point outside the shell, the shell behaves as if the whole of its mass is concentrated at the center of the shell.

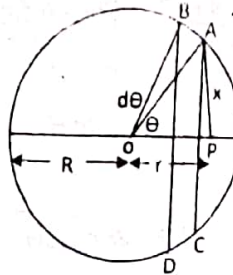
$$V \propto (1/r)$$

Gravitational field intensity,

$$E = -dV/dr$$

$$= -GM / r^2$$

(ii) At a point inside a shell:



Consider a uniform spherical shell of mass M and radius R . Let ρ be the density per unit area of the shell. The planes AC and BD cut the shell vertically and the element between the two planes is a ring of radius $R \sin \theta$.

So the length of AB ,

$$l = R d\theta$$

$$\text{Surface area of elements} = 2\pi R \sin \theta R d\theta$$

$$= 2\pi R^2 \sin \theta d\theta$$

Mass of the element,

$$m = (2\pi R^2 \sin \theta d\theta) \rho$$

If x is the distance of the point P from A , then the potential at P due to the element,

$$dV = - Gm/x$$

$$= - (G2\pi\rho R^2 \sin \theta d\theta) / x$$

From the triangle ΔOAP

$$x^2 = R^2 + r^2 - 2Rr \cos \theta$$

differentiating above equation we get,

$$x = (rR \sin \theta d\theta) / dx$$

So the potential,

$$dV = - (G2\pi\rho R^2 \sin \theta d\theta) dx / (rR \sin \theta d\theta)$$

$$= - (2\pi\rho GR / r) dx$$

Potential at a point inside the shell due to the element,

$$dV = - (2\pi\rho GR / r) dx$$

Now, to get the potential for the whole shell, integrate above expression between the limits $(R-r)$ and $(R+r)$

$$\begin{aligned} V &= - \int_{R-r}^{R+r} \frac{2\pi\rho GR}{r} dx \\ &= - G4\pi R\rho \\ &= - GM / R \end{aligned}$$

Thus, above equation shows that V is independent of r i.e; it is the same at all points inside the shell. This potential is equal to the potential on the surface of the shell and is constant.

Gravitational field intensity,

$$\begin{aligned} E &= -dV/dr \\ &= 0 \end{aligned}$$

Therefore, the gravitational field intensity at any point inside a spherical shell is zero.

Escape Velocity:

The minimum velocity with which an object must be thrown vertically upwards so that it does not return to the earth is called the escape velocity.

Expression for escape velocity:

If v is the escape velocity, then the initial kinetic energy of projection $(1/2)mv^2$ must be equal to the work done in moving the body from the surface of the earth to infinity. In such a case, the body does not return to the earth's surface.

Considering,

M is the mass of the earth, R is the radius of the earth and Universal gravitational constant is G .

Suppose a body of mass m is at a distance x from the earth's surface.

The gravitational force acting on the body is

$$F = GMm / x^2$$

✓ **Problem 1:** Determine the force of gravitational attraction between the earth of mass 5.98×10^{24} Kg and a 70 Kg boy who is standing at a sea level, a distance of 6.38×10^6 m from earth's center.

Solution:

Given that,

Mass of earth, $M = 5.98 \times 10^{24}$ Kg

Mass of boy, $m = 70$ Kg

Distance of boy, $r = 6.38 \times 10^6$ m

Gravitational constant, $G = 6.673 \times 10^{-11} \text{ Nm}^2\text{Kg}^{-2}$

We know that,

$$F = GMm / r^2$$

$$= 686.24 \text{ N}$$

✓ **Problem 2:** Determine the gravitational potential and field of a sphere at a distance 50mm from the center of gravity of the sphere of mass 50Kg and radius 10cm.

Solution:

Here,

Mass of the sphere, $M = 50$ Kg

Radius of the sphere, $R = 10\text{cm} = 0.1\text{m}$

Distance, $r = 50\text{mm} = 0.05\text{m}$

We know that,

The gravitational potential inside the sphere is equal to the potential on the surface of sphere.

So

Gravitational Potential,

$$V = -Gm/R = -3.34 \times 10^8 \text{ J Kg}^{-1}$$

Gravitational field intensity, $E = 0 \text{ N Kg}^{-1}$

✓ **Problem 3:** Calculate the escape velocity of a body from the moon's surface, if the radius of the moon is $1.7 \times 10^6 \text{ cm}$ and acceleration due to gravity on the moon's surface is 1.63 ms^{-2} .

Solution:

Here,

Radius of the moon, $R = 1.7 \times 10^6 \text{ cm} = 1.7 \times 10^6 \text{ m}$

Acceleration due to gravity on moon's surface, $g = 1.63 \text{ ms}^{-2}$

We know that,

$$V = \sqrt{2gR} = 2.35 \times 10^3 \text{ m/s} = 2.35 \text{ km/s}$$

✓ **Problem 4:** What will be the acceleration due to gravity on the surface of the moon if its radius were $\frac{1}{4}$ th radius of earth and mass $\frac{1}{80}$ th of the mass of earth? What will be the escape velocity on the surface of the moon if it is 11.2 km/s on the surface of the earth?

Solution:

Here,

Radius of the earth = R_e

Mass of the earth = M_e

Acceleration due gravity on earth's surface, $g_e = 9.8 \text{ m/s}^2$

Escape velocity of earth, $v_e = 11.2 \text{ Km/s} = 11200 \text{ m/s}$

So Radius of the earth, $R_m = \frac{1}{4} R_e$

$$\begin{aligned} V &= \sqrt{2gR} \\ \frac{V}{V_e} &= \sqrt{\frac{2gR}{2g_e R_e}} \\ \frac{V}{11200} &= \sqrt{\frac{gR}{g_e R_e}} \\ \frac{V}{11200} &= \sqrt{\frac{g \cdot \frac{1}{4} R_e}{9.8 R_e}} \\ \frac{V}{11200} &= \sqrt{\frac{g}{39.2}} \\ \frac{V}{11200} &= \frac{1}{2} \\ V &= 5600 \text{ m/s} \end{aligned}$$

Mass of the moon, $M_m = 1/80 M_e$

Escape velocity of moon, $v_m = ?$

We know that,

$$g_m = GM_m / R_m^2$$

$$\text{and } g_e = GM_e / R_e^2$$

so,

$$g_m / g_e = (M_m / M_e) \times (R_e / R_m)^2$$

$$\text{or, } g_m / g_e = 1/5$$

$$\text{or, } g_m = 1.96 \text{ m/s}^2$$

Again,

$$v_m = \sqrt{2 g_m R_m}$$

$$\text{And } v_e = \sqrt{2 g_e R_e}$$

Hence,

$$v_m / v_e = \sqrt{2 g_m R_m} / \sqrt{2 g_e R_e}$$

$$\text{or, } v_m = 2.50 \text{ km/s}$$

at the farthest point be 5×10^3 m/s, calculate the speed at the nearest point. [Lucknow, 1981]

According to Kepler's laws

$$\frac{1}{2} R^3 \omega = \text{const}$$

$$\frac{1}{2} R_1^3 \omega_1 = \frac{1}{2} R_2^3 \omega_2$$

or

$$\frac{1}{2} R_1 v_1 = \frac{1}{2} R_2 v_2$$

Here

$$R_1 = 1.5 \times 10^7 \text{ m}$$

$$R_2 = 0.5 \times 10^7 \text{ m}$$

$$v_1 = 5 \times 10^3 \text{ m/s}$$

$$v_2 = ?$$

$$v_2 = \left(\frac{R_1 v_1}{R_2} \right)$$

$$v_2 = \left(\frac{1.5 \times 10^7 \times 5 \times 10^3}{0.5 \times 10^7} \right)$$

$$v_2 = 1.5 \times 10^4 \text{ m/s.}$$

5.4 Newton's Universal Law of Gravitation

The motion of the planets round the sun led Newton to formulate the Universal Law of Gravitation. According to Newton, this law is true not only for the heavenly bodies but is also true for any two bodies in this universe. The law holds good for the force of attraction between the sun and the earth, for the attraction between the moon and the earth, for the attraction of an electron in the orbit and the nucleus in the atom etc.

According to this law :

Every body in this universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

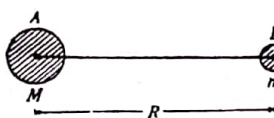


Fig. 5.22

Consider two bodies A and B of masses M and m respectively. The distance between their centres = R (Fig. 5.2)

$$F \propto Mm \quad \dots (i)$$

$$F \propto \frac{1}{R^2} \quad \dots (ii)$$

Combining (i) and (ii)

$$F \propto \frac{Mm}{R^2} \quad \text{or} \quad F = G \left[\frac{Mm}{R^2} \right]$$

Here G is the universal constant of gravitation.

If $M = m = 1$,

and

$$R = 1,$$

$$F = G$$

The constant of gravitation is numerically equal to the force of attraction between two unit masses at a unit distance apart.

$$\text{In CGS units} \quad G = 6.670 \times 10^{-8} \text{ dyne-cm}^2/\text{g}^2$$

In rationalised MKS unit,

$$G = 6.670 \times 10^{-11} \text{ newton-m}^2/\text{kg}^2$$

Example 5.1 Assuming the earth's orbit around the sun to be a circle, calculate the angular velocity of the earth about the sun. What is its average linear speed and centripetal acceleration with respect to the sun? Orbital radius of earth = 1.5×10^{11} m

$$\text{Here} \quad T = 365.25 \text{ days} = 365.25 \times 86400$$

$$T = 3.156 \times 10^7 \text{ s}$$

$$(i) \quad \omega = \frac{2\pi}{T} = \frac{2\pi}{3.156 \times 10^7}$$

$$\omega = 1.991 \times 10^{-7} \text{ radians/s}$$

$$(ii) \quad v = R\omega$$

$$v = 1.5 \times 10^{11} \times 1.991 \times 10^{-7}$$

$$v = 2.987 \times 10^4 \text{ m/s}$$

$$(iii) \text{ Centripetal acceleration}$$

$$= \frac{v^2}{R} = \frac{(2.987 \times 10^4)^2}{1.5 \times 10^{11}}$$

$$= 5.948 \times 10^{-3} \text{ m/s}^2$$

Example 5.2 Calculate the mass of the earth from the following data.

$$g = 9.80 \text{ m/s}^2$$

$$G = 6.67 \times 10^{-11} \text{ M.K.S units}$$

$$R = 6.38 \times 10^6 \text{ m}$$

$$F = \frac{GMm}{R^2}$$

$$mg = \frac{GMm}{R^2} \text{ or } g = \frac{GM}{R^2}$$

$$M = \frac{gR^2}{G} = \frac{9.80 \times (6.38 \times 10^6)^2}{6.67 \times 10^{-11}}$$

$$M = 5.98 \times 10^{24} \text{ kg}$$

Example 5.3 The period of revolution of Phobos around Mars is 0.319 d and its mean distance from the Mars is 9.519×10^6 m.

Calculate the mass of Mars (i) in kg and (ii) in units of earth's mass. Mass of the earth is 5.977×10^{24} kg.

Suppose mass of Mars = M

Mass of Phobos = m

$$M+m = \frac{4\pi^2 \left(\frac{a^3}{T^2}\right)}$$

Here, m can be neglected as compared to M

$$M = \frac{4\pi^2 \left(\frac{a^3}{T^2}\right)}$$

Here, $G = 6.66 \times 10^{-11}$ S.I. units

$$a = 9.519 \times 10^6 \text{ m}$$

$$T = 0.319 \text{ days} = 0.319 \times 86400 \text{ s}$$

$$M = \frac{4\pi^2 (9.519 \times 10^6)^3}{6.66 \times 10^{-11} (0.319 \times 86400)^2}$$

$$M = 67.33 \times 10^{23} \text{ kg}$$

$$\text{Mass of the earth} = 5.977 \times 10^{24} \text{ kg}$$

$$\text{Mass of the Mars} = \left(\frac{6.733 \times 10^{23}}{5.977 \times 10^{24}}\right) \text{ mass of the earth}$$

$$= 0.1127 \text{ times the mass of the earth}$$

Example 5.4 The mean distance of the moon from the earth is 3.844×10^8 m and its period of revolution around the earth is 27.32 days. Calculate the mass of the earth and the moon, if the centre of mass of the earth-moon system is 4.75×10^6 m from the earth's centre.

$$\text{Mass of the earth} = M_1$$

$$\text{Mass of the moon} = M_2$$

$$a = 3.844 \times 10^8 \text{ m}$$

$$T = 27.32 \times 86400 \text{ s}$$

$$G = 6.66 \times 10^{-11} \text{ S.I. units}$$

$$M_1 + M_2 = \frac{4\pi^2 a^3}{GT^2}$$

$$M_1 + M_2 = \frac{4\pi^2 (3.844 \times 10^8)^3}{6.66 \times 10^{-11} \times (27.32 \times 86400)^2}$$

$$M_1 + M_2 = 6.045 \times 10^{24} \text{ kg}$$

$$\text{also } a_1 = 4.75 \times 10^6 \text{ m}$$

$$a_1 + a_2 = a$$

$$a_1 + a_2 = 3.844 \times 10^8 \text{ m}$$

Gravity and Gravitation

$$a_2 = 3.844 \times 10^8 - 4.75 \times 10^6 \text{ m}$$

$$a_2 = 379.65 \times 10^6 \text{ m}$$

$$M_1 a_1 = M_2 a_2$$

$$\frac{M_1}{M_2} = \frac{a_2}{a_1}$$

$$\frac{M_1 + M_2}{M_1} = \frac{a_1 + a_2}{a_1} = \frac{a}{a_1}$$

$$M_2 = \frac{(M_1 + M_2) a_1}{a}$$

$$M_2 = \frac{6.045 \times 10^{24} \times 4.75 \times 10^6}{3.844 \times 10^8}$$

$$M_2 = 7.469 \times 10^{24} \text{ kg}$$

$$M_1 = 6.045 \times 10^{24} - 7.46 \times 10^{23}$$

$$= 5.97 \times 10^{24} \text{ kg.}$$

Example 5.4 (A). The maximum and minimum distances of a comet from the sun are 1.4×10^{13} m and 7×10^{12} m respectively. The velocity of the comet when it is nearest to the sun is 6×10^4 m/s. What will be its velocity when it is farthest from the sun, assuming that in both positions, the comet is moving in circular orbits? [Delhi, 1980]

In this case

$$\frac{GM}{r_1^3} = \frac{v_1^2}{r_1} \quad \dots(1)$$

and

$$\frac{GM}{r_2^3} = \frac{v_2^2}{r_2} \quad \dots(2)$$

Dividing (1) by (2)

$$\frac{r_2^3}{r_1^3} = \left(\frac{v_1^2}{v_2^2}\right) \left[\frac{r_2}{r_1}\right]$$

$$v_2^2 = v_1^2 \left(\frac{r_1}{r_2}\right)$$

$$v_2 = v_1 \left[\frac{r_1}{r_2}\right]^{\frac{1}{2}}$$

Hence

$$v_1 = 6 \times 10^4 \text{ m/s}$$

$$r_1 = 7 \times 10^{12} \text{ m}$$

$$r_2 = 1.4 \times 10^{13} \text{ m}$$

$$v_2 = (6 \times 10^4) \left[\frac{7 \times 10^{12}}{1.4 \times 10^{13}}\right]^{\frac{1}{2}}$$

$$v_2 = 1.311 \times 10^4 \text{ m/s}$$

5.9 Acceleration due to Gravity

Consider a body of mass m on the surface of the earth (Fig. 5.9). Mass of the earth = M and radius of the earth = R . The force of attraction exerted by the earth on the body according to Newton's law of gravitation.

$$F = \frac{GMm}{R^2}$$

According to Newton's second law of motion $F = ma$. Here $a = g$, where g is the acceleration due to gravity (Fig. 5.9).

$$F = ma$$

$$\text{or } mg = \frac{GMm}{R^2} \text{ or } g = \frac{GM}{R^2}$$

In CGS units, $g = 980 \text{ cm/s}^2$
In rationalised MKS units,

$$g = 9.80 \text{ m/s}^2$$



Fig 5.9

A body falling freely under the action of gravity towards the earth moves with a uniform acceleration equal to g .

Example 5.5 A body weighs 90 kg on the surface of the earth. How much will it weigh on the surface of the Mars whose mass is $1/9$ and radius $1/2$ that of the earth.

Mass of the earth = M , Mass of the body = m

Radius of the earth = R

From Newton's Law of Gravitation

$$F = \frac{G \times M \times m}{R^2}$$

$$F = 90 \text{ kg wt} = 90 \times 9.8 \text{ newtons.}$$

$$90 \times 9.8 = \frac{G \times M \times m}{R^2} \quad \dots(i)$$

$$\text{Mass of the Mars} = \frac{M}{9}, \text{ Radius of the Mars} = \frac{R}{2}$$

Suppose the wt. of the body on the surface of Mars = x

$$\text{Then } x = \frac{G \times \frac{M}{9} \times m}{\left(\frac{R}{2}\right)^2} \quad \dots(ii)$$

Dividing (ii) by (i)

$$\frac{x}{90 \times 9.8} = \frac{4}{9} \text{ or } x = \frac{90 \times 9.8 \times 4}{9} = 40 \times 9.8 \text{ newtons}$$

$$= 40 \text{ kg wt}$$

Thus, the body will weigh 40 kg wt on the surface of Mars.

Example 5.6 The mass of a planet is three times that of the earth and the diameter of the planet is also three times that of the earth. What is the acceleration due to gravity on the surface of the planet? Value of g on the earth's surface = 9.80 m/s^2

$$g = \frac{GM}{R^2} \quad \dots(i)$$

$$g' = \frac{GM'}{R'^2} \quad \dots(ii)$$

$$\therefore \frac{g'}{g} = \left(\frac{M'}{M}\right) \left(\frac{R^2}{R'^2}\right)$$

Here $\frac{M'}{M} = 3$

and $\frac{R}{R'} = \frac{1}{3}, g = 9.80 \text{ m/s}^2$

$$\therefore \frac{g'}{g} = 3 \times \frac{1}{9} = \frac{1}{3}$$

$$\therefore g' = \frac{g}{3} = \frac{9.8}{3} = 3.267 \text{ m/s}^2$$

Example 5.7 Calculate the radius of the moon's orbit assuming the orbit to be circular around the earth. Given $g = 9.8 \text{ m/s}^2$. Radius of the earth = $6.4 \times 10^6 \text{ m}$. Period of rotation of moon around the earth = 27.3 days. (Delhi, 1978)

Let the radius of the orbit be R and radius of the earth = r

$$\frac{g'}{g} = \frac{r^2}{R^2}$$

$$g' = \frac{gr^2}{R^2} \quad \dots(i)$$

Also

$$g' = \frac{v^2}{R}$$

$$\therefore \frac{gr^2}{R^2} = \frac{v^2}{R} = \frac{\left(\frac{2\pi R}{T}\right)^2}{R} = \frac{4\pi^2 R}{T^2}$$

$$R^3 = \frac{gr^2 T^2}{4\pi^2}$$

$$R = \left[\frac{gr^2 T^2}{4\pi^2}\right]^{1/3}$$

$$R = \left[\frac{9.8 \times (6.4 \times 10^6)^2 \times (27.3 \times 86400)^2}{4 \times (3.142)^2}\right]^{1/3}$$

$$R = 3.838 \times 10^8 \text{ m}$$